



Marihan Shehata

Nationality: Egyptian **Date of birth:** 15/11/1996 **Gender:** Female

Phone number: (+49) 015560153504

Email address: mirehan.m.sherif@mu.edu.eg

LinkedIn: [linkedin.com/in/marihan-shehata-bbbb79227](https://www.linkedin.com/in/marihan-shehata-bbbb79227)

Website: <https://www.researchgate.net/profile/Marihan-Shehata>

Website: <https://github.com/marihanshehata>

Home: 18 Johann Stelling Straße , 17489 Greifswald (Germany)

ABOUT ME

Research Assistant at the Computer and Systems Engineering Department, Minia University, with research experience in Machine Learning, Deep Learning, and Reinforcement Learning. My research focuses on reinforcement learning algorithms, hyperparameter optimization, and learning dynamics in autonomous driving applications. I am particularly interested in Deep Learning, Reinforcement Learning, AutoML/AutoRL, and Explainable AI (XAI), and I aim to pursue a PhD in these areas.

WORK EXPERIENCE

Department of Computer and Systems Engineering (CSE), Minia University – Minia, Egypt

University research assistant

[06/2020 – Current]

Research activities

- Conducted research on Deep Reinforcement Learning for autonomous driving systems
- Investigated hyperparameter sensitivity and its impact on reinforcement learning performance and learning dynamics.
- Worked on optimization and optimal control techniques for autonomous driving systems

Teaching activities

Teaching assistant for undergraduate courses:

- Machine Learning
- Programming
- Control Theory
- Systems Analysis
- Cyber Physical Systems

EDUCATION AND TRAINING

Master of Science (MSc) in Computer and Systems Engineering

Minia University, Egypt [06/2021 – 08/2026]

Field(s) of study: Computer and Systems Engineering (Deep Reinforcement Learning and Optimal Control) |

Thesis: Optimal Control Application in Motion Planning of Autonomous Vehicles

- Deep Reinforcement Learning for optimal control in autonomous driving systems
- Hyperparameter optimization and tuning in reinforcement learning
- Analysis of learning dynamics in RL systems

Bachelor of Science (BSc) in Computer and Systems Engineering

Minia University, Egypt [09/2014 – 07/2019]

Field(s) of study: computer and systems engineering | **Final grade:** Distinction with Honour

SKILLS

Python / Matlab/Simulink / Deep Reinforcement Learning / Machine Learning / Deep Learning / control systems / Data Science | Data Collection, Data Processing, Data Analysis, Data Visualisation / statistics / c Programming ,c++ programming / Linux / embedded systems

PUBLICATIONS

[2025]

[Hyperparameter Sensitivity Analysis Of Reinforcement Learning In Autonomous Driving Environments](#)

Journal Name: 11th International Conference on Natural Language Computing (NATL 2025) | Publisher: AIRCC

[2020]

[Optimal Control of Lane Keeping System Using Simulated Annealing and Linear Quadratic Regulator](#)

Journal Name: 15th International Computer Engineering Conference (ICENCO) | Publisher: IEEE

[2026]

Hyperparameter Regimes and Learning Dynamics in Deep Reinforcement Learning for Autonomous Driving s
ubmitted, under review

Journal Name: JES. Journal of Engineering Sciences

PROJECTS

BSc graduation project : Cyber Physical Emulator For Embedded Automotive Systems

- Developed a low-cost cyber-physical emulator for the testing and validation of embedded automotive control systems.
- Implemented Model-Based Design (MBD) methodologies to support the design, development, testing, and performance analysis of automotive controllers.

CERTIFICATIONS

[IBM , Coursera]

Python for Data Science, AI & Development

Link: <https://coursera.org/share/64c61459b229ef7da25961b4ae5b9d77>

[IBM, Coursera]

Data Analysis with Python

Link: <https://coursera.org/share/08372cbf0fe3ef9371981b80ad3b329e>

[DeepLearning.AI, Coursera]

Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization

Link: <https://coursera.org/share/2f0f8f4fb0f990d59c96b900396ba7ef>

[DeepLearning.AI, Coursera]

Neural Networks and Deep Learning

Link: <https://coursera.org/share/573d39a28b81960d2c7e8bb7ac95e3e8>

[IBM , Coursera]

Python Project for Data Science

Link: <https://coursera.org/share/38df304562f3e2c7c9a4fb584a1d0c04>

[IBM , Coursera]

Data Visualization with Python

Link: <https://coursera.org/share/3431ca817b1f9f7dd2ec207d672c34f0>

LANGUAGE SKILLS

Other language(s): English C1 | Deutsch A2